

verity

One server. Every drawing, job and test traced.

The complete technical description of the solution: platform specification, installable web application, server footprint and load, requirements, security, recovery, module-by-module design, AI options and the acceptance evidence DSEW receives at go-live.

PREPARED FOR

D & S Electric Works
109, Industrial Area Extension, Rau, Indore 453331

PREPARED BY

PlotArmour
team@plotarmour.in · +91 87440 69597

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CONFIDENTIAL DRAFT FOR FINAL REVIEW

Hardware figures for the Complete ERP with AI search are set in section 7. Figures stated as targets are confirmed at the pre-deployment survey and proven by the acceptance tests in section 15. Companion documents: 1 Proposal · 2 Delivery and deployment · 3 Licence and ownership · 4 Company and platform.

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HOW TO READ THIS DOCUMENT

Sections 1 to 3 give the shape of the solution. Sections 5 to 7 answer the question DSEW's IT team will ask first: what does this ask of our server, and will it slow anything else down. Sections 4 and 8 to 12 are the platform and module specification. Sections 13 to 17 describe what is optional, how it is proven, and how it is supported.

SECTION 1

Summary for DSEW management and IT

Verity is a modular business operating platform. For DSEW it is configured as an engineering and manufacturing ERP that traces each engineered, one-off product from enquiry through approved drawing, bill of materials, manufacture and testing to dispatch. It runs on one Linux virtual machine on DSEW's own premises, with nothing leaving DSEW's network by default.

1 VM

One Linux virtual machine hosts the whole solution. No cluster, no cloud dependency

4 vCPU · 16 GB

Core ERP sizing for 15 named users. 32 GB when AI semantic search is enabled

0 licences

No paid third-party software. PostgreSQL, Keycloak, Next.js and the rest are open source

Any device

Installable web app (PWA) for office PCs, shop-floor phones and tablets

Thin client

All processing is on the server. User devices only display and capture

19 modules

On one shared foundation in the Complete ERP. 9 in the MVP

What DSEW gets, in technical terms

Application	One web application (TypeScript, Next.js, React) served over HTTPS, installable on any device as a Progressive Web App
Database	PostgreSQL 16, with database-enforced data isolation, a least-privilege application account and an append-only audit trail
Sign-in	Keycloak identity provider on the same VM; password policy, optional multi-factor sign-in, session control
Files	S3-compatible object storage on a separate volume for drawings, certificates, test reports and photographs
Packaging	Containerised with scripted install, backup, restore, upgrade and rollback
Controls	Role-based permissions, approval workflows, revision control, quality gates that block release, immutable audit
Server load	Designed to a stated resource budget, protected by caching, pagination, indexing, background-job scheduling and rate limits, and proven by a load test on DSEW's own VM before go-live (sections 6 and 15)
AI	Optional tiers by hardware, from rule-based detection and full-text search on the same VM up to a separate on-premises GPU machine. AI never approves, releases or changes a record (section 13)

Ownership	DSEW owns its data and can export it at any time. Source code of the DSEW-specific deliverables is handed over on full payment (document 3)

THE SERVER-LOAD ANSWER IN ONE PARAGRAPH

Verity is not a heavy system. It is one application process and one database, sized for a four-core, 16 GB VM and not a data centre. User devices carry no processing load. Report generation, file indexing and any AI indexing run as scheduled background work outside office hours, with limits so that a heavy job can never starve interactive users. DSEW's other workloads on the same Hyper-V host are protected by fixed CPU and memory allocation to the VM. A load test against agreed response-time targets on DSEW's actual VM is a go-live gate.

SECTION 2

Design principles for a customer-hosted system

Most business software is built to run in a vendor's cloud, where capacity can be added at will. A system that runs on the customer's own server has the opposite constraint: the hardware is fixed, it is shared with other work, and the customer's IT team carries the operational risk. Verity's deployment design for DSEW follows eight principles.

#	PRINCIPLE	WHAT IT MEANS FOR DSEW'S SERVER
1	Small, known footprint	A fixed set of five services on one VM, each with a memory budget (section 6). There is no message-broker cluster, no search cluster and no second database engine to feed.
2	Server does the work, device displays	Office PCs, phones and tablets render pages and capture input. Nothing computational is pushed to them, so old or low-end devices work.
3	Interactive work first	Anything slow (reports, exports, indexing, backups, AI) runs as scheduled background work with concurrency limits and runs outside shift hours by default.
4	Bounded queries	Every list is paginated and indexed. No screen loads an unbounded table. Search is database-indexed.
5	Fail closed	If tenant context or permission is missing, queries return nothing and writes are rejected. A fault never widens access.
6	Everything can be put back	Scripted backup, tested restore, versioned upgrades with rollback. Recovery is rehearsed before go-live, not merely described.
7	Data stays on the premises	No cloud service is required to run, and none receives DSEW data by default. Outbound access is needed only for install and update, and can be replaced by offline media.
8	Standardise the foundation, configure the behaviour	Identity, permissions, audit, numbering, workflow and storage are one shared foundation. DSEW's specific behaviour is configuration and modules on top, so later additions do not require redevelopment.

SECTION 3

Solution architecture

Verity is built as layers. Each layer has one job, and an upper layer cannot bypass the one below it. This is what allows new capabilities to be added later without redesigning the foundation.

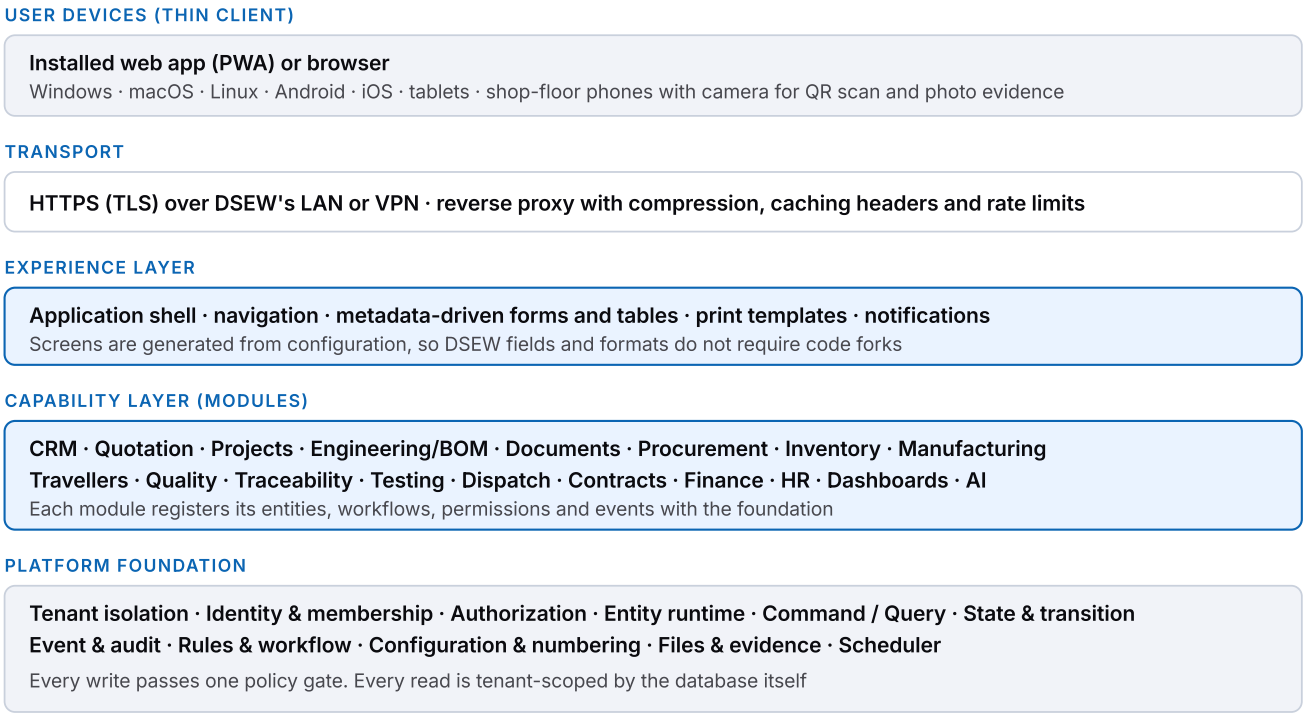


Figure 1. Layers of the solution. Modules are added in the capability layer without changing the foundation.

3.1 Runtime components

COMPONENT	TECHNOLOGY	ROLE
Application	TypeScript, Next.js, React	Serves the interface and the permissioned API. Runs the business logic: commands, queries, state transitions, rules and events
Database	PostgreSQL 16	System of record. Row-level security enforces company isolation. Holds records, audit trail, the event log and full-text indexes
Identity	Keycloak (OpenID Connect)	Sign-in, password policy, optional multi-factor sign-in, session lifetime. Application roles remain in Verity
File storage	S3-compatible object store	Drawings, certificates, reports, photographs. Held on a separate volume so file growth never competes with the database
Reverse proxy	Standard TLS-terminating proxy	HTTPS, compression, static-asset caching, rate limiting, security headers
Packaging	Docker and Compose	Reproducible install; versioned images; scripted upgrade and rollback

COMPONENT	TECHNOLOGY	ROLE
Scheduler	Built in, one trigger	Runs background work (reminders, escalations, reports, indexing, backups) on a schedule with concurrency limits

3.2 Path of a single request

1. The user's device sends an HTTPS request carrying a signed session issued after Keycloak sign-in.
2. The application derives the **company (tenant) context from the authenticated session**, never from anything the browser sends in the request body.
3. For a change, the request becomes a **command**. One policy gate checks the user's permission (action, entity, scope) and the record's current state.
4. Inside one database transaction the tenant is set, the state change is written, the **audit entry and event** are recorded, and the transaction commits or rolls back as a whole.
5. For a read, a **query** returns only the rows that tenant and scope permit; the database applies the same filter even if the application code were wrong.
6. The response returns a small payload. Large lists are paged; files are streamed from the object store.

3.3 Extension model

A new capability is registered, not patched in. It declares its entities, workflows, permissions, events, screens and the other capabilities it depends on. DSEW's later additions (a new product family, an inspection format, a new department) are configuration or added modules; the foundation, the workflow engine and the authorisation model are not modified, and no code is forked per client.

SECTION 4

Verity platform specification

This section defines the platform foundation on which all 19 modules rest. These are the building blocks that make the system consistent, auditable and extensible.

4.1 Identity and data model

CONCEPT	DEFINITION
Tenant	The root data-isolation boundary. DSEW is one tenant. Every record belongs to exactly one tenant and no record references another tenant's data
Organization	A business unit inside the tenant (for example a plant or a department). Permissions can be scoped to it
Party	One record per person or business: customer, supplier, employee, contact. A single identity, never duplicated across modules
User	A person who signs in. One-to-one with a Party
Membership	A user's place in an organization, which carries role assignments
Location	A site, store or bay. Distinct from an address and from a geofence
Resource	A schedulable unit backed by a person or an asset: a machine, a tester, a technician
Asset	Physical equipment: machines, test instruments, tools
Work	A unit of work to be executed: a manufacturing job, an inspection, a project task
Evidence	Immutable field data: photographs, measured values, signatures, scanned documents
Contract	An agreement with its terms, obligations and exposures

4.2 Authorization

- **Permissions are Verb + Entity + Scope.** For example *approve · Quotation · Organization* or *release · BillOfMaterials · Tenant*. There are no hard-coded roles such as "manager" in the code; roles are named sets of permissions that DSEW defines.
- **Scopes** narrow where a permission applies: tenant-wide, one organization, or one location. Additional axes such as team or resource are handled by configuration.
- **One decision point.** Every command, from the screen, the API, a scheduled job or an AI assistant, passes the same policy check. There is no back door and no elevated service path.
- **Separation of duties** is configurable: DSEW chooses who drafts, reviews, approves and releases, and whether self-approval is ever allowed.

- **The AI channel acts as the calling user.** Whatever an AI feature does, it does with that user's own permissions and nothing more (section 13).

4.3 State and transition

Business records move through configurable **state machines**. Every state belongs to one of six behavioural categories, so reports and timers can read behaviour without depending on DSEW's own labels.

CATEGORY	MEANING	EXAMPLE DSEW STATES
Draft	Being prepared; not yet in force	Quotation draft, BOM draft
Pending	Waiting for a decision or input	Awaiting approval, awaiting material
Active	In progress	In winding, in assembly, in testing
Blocked	Cannot proceed until a condition clears	On hold for NCR, awaiting drawing revision
Completed	Execution finished (terminal)	Tested and passed
Cancelled	Abandoned (terminal)	Enquiry lost, order cancelled

Completed and Closed are different. Completed means the work is done. Closed is an administrative step that makes the record **permanently read-only**. A closed record cannot be edited; later rework creates a linked child record. This is how "what was approved is what was built" is enforced by the system and not by habit.

4.4 Commands, queries, events and audit

Command	A request to do something: create a quotation, release a BOM, issue material. Validated, authorised, state-checked and applied in one transaction
Query	A read of current state, always tenant-scoped and permission-scoped
Event	A fact that something happened (*BOM released*, *inspection failed*). Other modules and notifications react to events, so modules stay loosely coupled
Projection	A derived read view (dashboards, ageing, stock balance) built from events and records
Audit trail	For every change: who, what, when, from which channel, previous and new values. Written in the same transaction as the change; the application cannot alter or delete it
Failed attempts	Refused commands are recorded as metadata, so repeated denied attempts are visible to the administrator

4.5 Rules, workflow and configuration

- **Workflow definitions** are data: nodes (steps), edges (transitions) and conditions (for example "approval required above a value", "all mandatory checks passed").
- **Custom fields** add DSEW-specific attributes (kVA rating, impedance, cooling class, customer drawing number) to any record without schema forks.
- **Document numbering** is configurable per document type with prefix, year, site and sequence, allocated safely under concurrent use so numbers are never duplicated.
- **Forms and tables** are generated from metadata, so adding a field or a column is configuration with permission, validation and audit already applied.
- **Print templates** produce DSEW's branded quotation, traveller, challan, packing list, certificate and report formats.
- **Capability activation**: modules are switched on per tenant, and version pins mean DSEW upgrades on its own schedule.
- **Nothing DSEW-specific is hard-coded.** Test templates, inspection checklists, approval workflows, manufacturing routes, teams, roles, report and print formats, notification rules, categories and master data are settings held as data and edited by DSEW's administrators, with permission and audit applied. The fixed foundation (tenant isolation, sign-in, the permission check, the audit trail, the closed-record rule) is not configurable by design. The full list and its boundary are in document 1, section 2.

4.6 Application programming interface

A permissioned REST API exposes the same commands and queries the screens use, under the same policy gate. DSEW can read its data programmatically for reporting. No external system integration is included in scope (document 1, section 4).

SECTION 5

Installable web application (PWA) and device reach

Verity is delivered as a Progressive Web App. It opens in any modern browser, and on any device it can also be installed with one tap so it behaves like a native application, with no app store, no per-device installer and no separate mobile build to maintain.

5.1 Why this matters for DSEW

One build, every device

The same application serves office PCs, tablets and shop-floor phones. There is no Android build, iOS build and Windows build to keep in step, so every user sees the same version on the day it is upgraded.

Zero client installation work

Users open an internal address and choose "Install". IT does not push software to devices. Removing it is just as simple, and a lost phone holds nothing once its session is revoked.

Light on the device

Pages are server-rendered and compressed, so a modest phone or an older office PC performs well. The installed app is a shortcut to the same web application, not a second copy of the system.

Light on the server

The application shell, fonts, icons and scripts are cached on each device after the first visit. Later visits fetch only data, which cuts repeated traffic and CPU work on DSEW's server.

5.2 What is delivered

CAPABILITY	SPECIFICATION
Web app manifest	Name, icons for all sizes, theme colour and standalone display, so the installed app opens in its own window without browser chrome
Service worker	Caches the application shell and static assets. Business data is never served stale from cache for a transaction; it is always read from the server
Secure by default	PWA features require HTTPS. The internal TLS certificate arranged in the pre-deployment check (document 2) satisfies this
Camera capture	Photograph upload for inspection evidence and QR scanning with the device camera. Keyboard-style hardware scanners also work as ordinary input
Responsive layouts	Phone, tablet and desktop layouts. Touch targets are sized for shop-floor use. Tables adapt to narrow screens

CAPABILITY	SPECIFICATION
Update behaviour	When DSEW upgrades the server, the next load picks up the new version and the user is prompted to refresh. No device-by-device update
Connection state	If the network drops, the application shows a clear "reconnecting" state and protects unsaved entries from being lost. It does not silently queue business transactions that could conflict
Accessibility	Keyboard operation, visible focus, screen-reader labels, and text contrast checked to WCAG AA in light and dark appearance

5.3 Supported devices

DEVICE	BROWSERS	INSTALL METHOD	TYPICAL DSEW USE
Windows PC / laptop	Chrome, Edge, Firefox	Install icon in the address bar (Chrome, Edge)	Office, engineering, stores, accounts
macOS / Linux	Chrome, Edge, Firefox, Safari (macOS)	Install from the browser menu	Engineering, management
Android phone / tablet	Chrome, Edge	"Install app" prompt	Shop floor, inspection, stores, QR scan
iPhone / iPad	Safari	Share → Add to Home Screen	Management, shop floor, inspection
Thin client / kiosk	Chrome or Edge, full-screen	Shortcut to the internal address	Shared terminal at a workstation

5.4 Network use on the premises

In-LAN use needs no internet after installation. Shop-floor phones reach the server over the premises Wi-Fi to the internal HTTPS address. Remote use goes only through DSEW's VPN. Page weight after the first load is small, because the shell is cached and responses are compressed, so ordinary Wi-Fi is sufficient.

SECTION 6

Server load and performance engineering

DSEW will host Verity on its own server, so the load Verity places on that server is a design requirement, not an afterthought. This section sets out where load comes from, how it is held down, how much of the VM each part may use, and how it is proven before go-live.

6.1 Where load comes from, and how each source is controlled

SOURCE OF LOAD	WHY IT CAN BE HEAVY	CONTROL BUILT INTO THE SOLUTION
Interactive screens	Large lists and joins across modules	Server-side pagination, indexed queries, bounded page size, selective columns. No screen loads an unbounded table
Search	Scanning free text across records and documents	PostgreSQL full-text indexes maintained on write; search never scans tables
Reports and dashboards	Aggregations over many records	Precomputed projections refreshed in the background; heavy reports queued to run off-hours; result caching
File upload and download	Large drawings and scans	Streamed to and from the object store, not buffered in memory; size limits per type; separate volume
Background jobs	Reminders, escalations, indexing, exports	One scheduler, concurrency caps, run windows outside shift hours, jobs yield to interactive traffic
Concurrency spikes	Many users at shift start or approval deadlines	Database connection pooling in transaction mode; request rate limits; short transactions
Backups	Disk and CPU during dump	Scheduled outside working hours, low priority, written to the separate backup target
AI indexing (optional)	Embedding generation is CPU-intensive	Background only, throttled, paused when interactive load is high (section 13)

6.2 Resource budget for the core VM (4 vCPU, 16 GB)

The VM is given fixed resources so it cannot grow into the memory or CPU that DSEW's other workloads on the Hyper-V host need, and the budget below leaves headroom inside the VM so that peaks do not push it into swapping.

SERVICE	PLANNED MEMORY	NOTES
Operating system and tools	≈ 1.0 GB	Ubuntu Server LTS, no desktop environment

SERVICE	PLANNED MEMORY	NOTES
Application (Node runtime)	≈ 1.5 to 2.5 GB	Worker count tuned to the cores; bounded heap
PostgreSQL 16	≈ 3.0 to 4.0 GB	Shared buffers plus working memory, bounded per connection
Keycloak	≈ 1.0 to 1.5 GB	Authentication only; light for 15 users
Object store and proxy	≈ 0.5 to 1.0 GB	Stream-oriented
Planned total in use	**≈ 7 to 10 GB**	Remaining memory serves as the operating system's file cache, which speeds up database reads

PLANNING FIGURES

The values above are the design budget used to size the VM. The actual footprint is measured on DSEW's VM during installation and recorded in the acceptance report (section 15).

6.3 Workload model used for sizing

PARAMETER	PLANNING VALUE
Named users	Up to 15 (scales with the sizing table in 6.4)
Concurrent active users	Up to 10 to 12 at peak, since not all named users are active together
Typical interaction	One page action every 20 to 40 seconds per active user; most are reads
Peak events	Shift start (many sign-ins and dashboard views); month-end (reports); approval cut-offs
Document volume	Drawings, certificates, reports and photographs attached to projects and jobs (section 8)

6.4 Sizing by users and data

SCENARIO	VCPU	MEMORY	DISK	NOTES
Up to 15 users, core ERP	4	16 GB	Within the 500 GB SSD	The offered configuration
Up to 15 users, with AI semantic search	4 to 6	32 GB	Within the 500 GB SSD, plus index storage	Embedding model and vector index compete for memory
Up to 30 users	6	24 to 32 GB	As above	Quoted separately; no redesign needed

SCENARIO	VCPU	MEMORY	DISK	NOTES
Up to 60 users	8	32 GB	Larger SSD	Quoted separately; same architecture
Interactive AI drafting	n/a	n/a	n/a	Separate on-premises GPU machine on the LAN (section 13); the ERP VM is not resized for it

6.5 Response-time targets proven at acceptance

These are the targets the load test is run against on DSEW's VM at 15 concurrent sessions with realistic mixed actions. They are agreed with DSEW at requirements sign-off and the results are recorded in the acceptance report.

ACTION	TARGET (95TH PERCENTILE)
Open a list page of up to 50 rows (filtered, sorted)	under 1.5 s
Open a record with its linked data	under 1.5 s
Save a record or transition a state	under 1.0 s
Full-text search across records and documents	under 2.0 s
Management dashboard load	under 3.0 s
Upload of a 10 MB file over the LAN	under 5 s
Sustained VM CPU during the test	average under 60 %
Memory in use, no swapping	within the planned budget

6.6 Protecting DSEW's other workloads

- **Fixed allocation:** the VM is created with static memory and a defined number of virtual processors, so Verity cannot take more of the host than agreed.
- **Disk separation:** database, files and backups are on separate volumes, so file growth or a backup cannot starve database input and output.
- **Quiet background:** backups, indexing and large reports run in a scheduled window that DSEW sets, at low priority.
- **Visibility:** a health view for the administrator shows CPU, memory, disk, database size, queue depth and slow queries, with alert thresholds, so capacity is a managed number and never a surprise.
- **Growth path:** if DSEW outgrows the VM, resources are increased on the same architecture. There is no re-platforming.

SECTION 7

Minimum and recommended requirements

Final figures are confirmed in the pre-deployment survey (document 2, section 6). Where this section differs from the hardware line in document 2 for the Complete ERP with AI search, this section governs.

7.1 Server (one virtual machine)

ITEM	MINIMUM (CORE ERP, 15 USERS)	RECOMMENDED	COMPLETE ERP WITH AI SEARCH
Virtual processors	4	4 to 6	4 to 6
Memory	16 GB, static	16 GB	**32 GB**, static
System and database disk	500 GB SSD	500 GB mirrored SSD	500 GB mirrored SSD plus index space
Document volume	Separate volume, sized with DSEW	Separate volume	Separate volume
Operating system	Ubuntu Server LTS (Linux guest). Not run natively on Windows Server and not run through Docker Desktop or WSL		
Host	Hyper-V role enabled with hardware virtualisation on, or a small dedicated host		

7.2 Network and security services

ITEM	REQUIREMENT
Address and name	A fixed IP address and an internal DNS name for the VM
LAN	Gigabit LAN between the server and office switches; reliable Wi-Fi coverage on the shop floor
TLS certificate	A certificate for the internal name, trusted by office PCs and phones (public CA or DSEW's internal CA). Required for sign-in security and for the installable web app
Remote access	DSEW's VPN, only if remote use is wanted
Internet	Controlled outbound access for installation and updates, or offline media. None at runtime
Time	Time synchronisation (NTP) so audit timestamps are correct

7.3 Backup and power

ITEM	REQUIREMENT
Backup target	A separate device or share, at least equal to the data size, ideally with a second off-site or offline copy
Power	UPS sized for a clean shutdown of the host and the VM

7.4 User devices

ITEM	REQUIREMENT
Office PC	Any PC from the last several years with a current Chrome, Edge or Firefox. No local software installation required
Phone / tablet	Android 9 or later with Chrome, or iOS 16 or later with Safari; rear camera for photographs and QR scan
Screen	1366 × 768 or larger recommended on desktop; any phone or tablet size is supported

7.5 People

A named DSEW product owner (about 3 to 4 hours a week), leads for quality, engineering, stores and finance, and a named IT contact with the rights needed during the survey and installation (document 2, section 12).

7.6 Optional AI hardware

Only if DSEW chooses interactive AI drafting: a separate small workstation with a 16 GB GPU on the LAN. It is not part of the VM and not included in the fee. Details and tiers are in section 13.

SECTION 8

Data architecture and storage growth

8.1 Data domains

DOMAIN	MAIN RECORDS
Identity and access	Tenant, Organization, Party, User, Membership, Role, Permission, Grant
Commercial	Customer, Contact, Enquiry, RFQ, Quotation and revisions, Sales Order, Contract, Retention, Liquidated-damages terms
Project	Project, Milestone, Task, Budget, Risk, Change request
Engineering	Product, Component, BOM and revision, Engineering Change Request and Order, Drawing
Documents	Controlled document, Revision, Transmittal, Acknowledgement, Numbering series
Supply chain	Supplier, Purchase Request, RFQ, Purchase Order, Goods Receipt, Stock ledger, Reservation, Warehouse and bin
Manufacturing	Manufacturing job, Operation, Route, Traveller, Work centre, Labour, WIP, Subcontract
Quality and test	Inspection plan, Checklist, Result, NCR, CAPA, Calibration, QAP/ITP, Test record, Certificate
Traceability	Batch, Heat, Serial, QR identity, Genealogy link
Logistics	Dispatch plan, Packing list, Challan, Shipment
Finance and HR	Invoice, Payment, Receivable, Payable, Profitability; Employee, Attendance, Leave, Payroll run, Payslip
Platform	Audit trail, Event log, Notification, Scheduled job, Configuration

8.2 Where storage growth actually comes from

Structured records are small. Storage growth is dominated by **files** (drawings, scans, photographs, test reports), which is why files live on their own volume and not inside the database.

STORE	GROWTH DRIVER	PLANNING BASIS
Database (records, audit, events, search indexes)	Transactions and revisions	Tens of thousands of records a year at DSEW's volume; a few gigabytes over five years including indexes and audit
Document volume	Files per project and job	Sized with DSEW from its sample documents; typically the largest consumer
Vector index (AI semantic search only)	Chunks of indexed documents	Proportional to indexed text, not to image or CAD file size

STORE	GROWTH DRIVER	PLANNING BASIS
Backups	Retention policy	At least equal to data size on the backup target, plus retention copies

PLANNING FIGURES

Growth values are planning assumptions from typical engineering-order volumes. The document-volume size is finalised in the survey from DSEW's real samples, and a capacity report is part of handover.

8.3 Data lifecycle and export

- **Retention:** closed records stay readable and locked; retention periods for documents are configured per type.
- **Export:** a documented export of all records as CSV and all files in original form with an index, at handover and once on request during the testing period.
- **Data location:** all data is held on DSEW's server. Support access is audited and time-limited.
- **Deletion:** at the end of an engagement, PlotArmour deletes or returns any DSEW material it holds.

SECTION 9

Security and data protection

Security is part of the foundation. It is applied to every operation from the first release, not added as a layer afterwards.

9.1 Controls by layer

LAYER	CONTROL
Transport	TLS for all traffic; HTTP redirected to HTTPS; modern protocol versions only; security headers (content security policy, frame protection, referrer policy)
Authentication	Keycloak with password policy and lockout; optional multi-factor sign-in, enabled on request; configurable session lifetime and idle timeout; immediate session revocation
Authorization	Verb + Entity + Scope permissions checked on the server for every command; no reliance on hidden screens; one policy gate for all channels
Tenant isolation	Company context comes from the authenticated session, never from the request. Row-level security in PostgreSQL enforces isolation even if application code were wrong; with no scope set, queries return nothing and writes are rejected
Database account	The application connects as a least-privilege account that cannot bypass row-level security. Start-up refuses a privileged account. A separate account is used only for schema migration
Audit	Append-only trail of every change with actor, time, channel and before/after values; cannot be edited or deleted by the application
Input handling	Server-side validation; parameterised queries; output encoding; protection against cross-site request forgery; upload type and size checks
Secrets	Held in a protected environment file (restricted permissions, owned by a named administrator), never in the database or in source. A sealed offline copy is held by DSEW's owner
Files	Access through the application's permission check; files are not publicly addressable
Backups	Encrypted at rest on the backup target; restore tested
Disk	Full-disk encryption recommended on the VM and backup target, confirmed at acceptance
Network exposure	Only the HTTPS port is exposed. Database, identity and object store are bound to the VM's internal network and not reachable from the LAN
Software supply	Pinned, versioned releases; dependency vulnerability review at each release; tested upgrade and rollback
Support access	Remote, time-limited, audited, and only with DSEW's permission

9.2 Rule for every change to data

For each state-changing operation the design records: who is acting, the permission required, the scope, the preconditions, the state change, the audit entry, the event emitted, and the behaviour on failure. A change that cannot answer all eight is not released.

9.3 AI and data protection

Nothing leaves DSEW's server to an external AI service unless DSEW explicitly enables that service. Local AI features run inside the signed-in user's own permissions, so a user can never retrieve through AI what they could not open directly. AI is never an approver and never changes a record on its own (section 13).

9.4 Scope boundary

Independent penetration testing and statutory certification are outside this scope. They can be arranged separately.

SECTION 10

Availability, backup, recovery and upgrades

10.1 Backup design

COMPONENT	METHOD	FREQUENCY (DEFAULT, AGREED WITH DSEW)
Database	Consistent logical dump; optional continuous log archiving for finer recovery	Nightly full; optional archiving for a shorter data-loss window
Files	Incremental copy of the document volume	Nightly
Identity configuration	Realm export	Nightly and before every change
Application configuration	Environment and version record	With every upgrade
Retention	Daily, weekly and monthly generations	Set with DSEW

10.2 Recovery objectives (proposed, agreed with DSEW)

OBJECTIVE	DEFAULT TARGET	TIGHTER OPTION
Data-loss window (RPO)	Up to 24 hours (nightly backup)	Up to 15 minutes with log archiving
Time to restore (RTO)	Within 4 hours on the same or a replacement VM	Agreed after the restore rehearsal

A **combined restore** of database, files and identity configuration onto a clean environment is rehearsed during the project (after Phase 2) and again as a final test before go-live. The restore time measured in the rehearsal replaces the default target in the handover documents.

10.3 Failure behaviour

FAILURE	BEHAVIOUR
Identity service unavailable	Existing sessions continue; new sign-ins fail with a clear message; quality gates remain enforced
Object store unavailable	Records remain usable; new uploads fail clearly; no record claims a file that was not stored
Application restart	Short interruption; no partial writes because every change is one transaction

FAILURE	BEHAVIOUR
Power loss	Database recovers to the last committed transaction on restart; UPS provides clean shutdown
Disk filling	Alerts at configured thresholds before the limit; document growth is on a separate volume
Network interruption on a device	The application shows connection state and protects unsaved input

10.4 Upgrades and rollback

1. Release notes and the changes to DSEW's installation are shared in advance.
2. A backup is taken automatically before the upgrade.
3. The upgrade is applied with database migrations in a defined order.
4. Health checks confirm the new version. If they fail, the scripted **rollback** restores the previous version and data state.
5. Upgrades are applied at a time DSEW chooses, typically outside working hours.

SECTION 11

Module-by-module technical specification

All modules share the foundation in section 4, so each inherits permissions, audit, numbering, search and event notifications. The table lists the main records, workflow states, controls and links to other modules for the Complete ERP. The MVP includes the basic layers described in document 1, sections 3 and 5.

#	MODULE	MAIN RECORDS AND WORKFLOW	CONTROLS AND RULES	LINKS TO
1	CRM, enquiry, quotation, sales order	Customer, contact, enquiry, RFQ, tender, quotation with revisions, sales order. Draft → Pending approval → Active → Won / Lost	Revision history; approval above configured value; alternative offers; PO comparison; retention and LD terms captured; follow-up reminders	Project, BOM, finance, documents
2	Advanced project management	Project, work breakdown, milestone, task, budget, risk, change. Draft → Active → Blocked → Completed → Closed	Dependencies; budget versus actual; change control; status roll-up	Orders, manufacturing, finance, contracts
3	Engineering and product lifecycle	Product, component, multi-level BOM and revision, ECR and ECO. Draft → In review → Released → Superseded	A released revision cannot be edited; every order records its revision; changes follow ECR/ECO approval	Manufacturing, procurement, travellers
4	Document management and controlled workflows	Controlled document, revision, transmittal, acknowledgement. Draft → Review → Approved → Issued → Obsolete	Configurable numbering; drafter, reviewer, approver and releaser roles; full revision and distribution history; search by number, title, content	All modules
5	Manufacturing execution	Manufacturing job, operation, route stage, work centre. Planned → In progress → On hold → Completed	Job identifies its BOM; material availability shown; operations tracked; hold and resume with reason; WIP; subcontract tracking	BOM, inventory, traveller, quality
6	Travellers and process documentation	Traveller generated from job and route; checkpoints and sign-offs	BOM-driven content; QR code on printout; sign-off by role; customer-facing formats	Manufacturing, quality, traceability
7	Quality management system	Inspection plan, checklist, result, NCR, CAPA, calibration, audit, QAP/ITP	Mandatory checks block release; NCR closes only after CAPA verification; evidence attached to results	Manufacturing, dispatch, testing, suppliers

#	MODULE	MAIN RECORDS AND WORKFLOW	CONTROLS AND RULES	LINKS TO
8	Barcode, QR and traceability	Batch, heat, serial, QR identity, genealogy	Unit-to-component genealogy; scan to open the record; QR verification page for a finished unit	Inventory, manufacturing, quality, dispatch
9	Procurement and vendor management	Supplier, purchase request, RFQ, comparative statement, PO, receipt. Draft → Approved → Ordered → Received	Approval workflow by value; supplier performance; subcontract orders; match to receipt	Inventory, finance, quality
10	Inventory and warehouse	Item, stock ledger, reservation, warehouse and bin, count	Ledger with reversal-only corrections (no silent edits); valuation; reconciliation; ageing; batch/heat/serial	Procurement, manufacturing, finance
11	Production planning and basic MRP	Demand, supply, planned order, shortage	BOM-based requirements; shortage recommendations; capacity view; scheduling by work centre	BOM, inventory, procurement, manufacturing
12	Testing and equipment	Test template, test record, certificate, instrument, calibration, maintenance	Tests per product family with acceptance limits; automated report; failed test blocks dispatch; calibration-due alerts	Quality, dispatch, assets
13	Finance and accounting	Invoice, payment, receivable, payable, GST rates, profitability	Invoices from orders and milestones; payments reduce balances; receivables reconcile; job profitability reconciles to costs	Sales, procurement, projects, payroll
14	HR and payroll	Employee, attendance, leave, shift, timesheet, salary structure, payroll run, payslip	Payroll from DSEW-supplied statutory components; a sample month reproduces DSEW's own calculation; competency tracking	Projects, manufacturing labour
15	Dispatch and logistics	Dispatch plan, packing list, challan, shipment. Ready → Packed → Dispatched	Dispatch released only after quality approval and test pass; partial deliveries; shipment tracking	Quality, inventory, finance
16	Contracts, compliance and risk	Contract, obligation, bank guarantee, LD exposure, insurance, NDA, risk register	Obligation due-date alerts; LD exposure tracking; expiry reminders	Orders, projects, finance
17	Dashboards and business intelligence	Role-based dashboards, KPI views	Precomputed projections; profitability, ageing, cash flow, supplier and resource analytics; respects permissions	All modules

#	MODULE	MAIN RECORDS AND WORKFLOW	CONTROLS AND RULES	LINKS TO
18	Notifications and workflow automation	Rules, triggers, escalations, templates	Event-driven; in-app and e-mail through DSEW's mail server; escalation chains; automated documents	All modules
19	AI (documented use cases)	Search, retrieval, classification, gap detection, summaries, drafts	Inside the caller's permissions; never an approver; acceptance threshold per use case on agreed samples	Documents, quotation, quality

SECTION 12

Engineering, manufacturing and quality controls

DSEW builds engineered, one-off products. The central requirement is that what was approved is what is built, and nothing leaves the factory without its checks. Verity enforces this through five connected controls.

12.1 From enquiry to dispatch: the controlled chain



Amber steps are gates: they block the next step until the condition is met

Figure 2. Linked records from enquiry to dispatch, with two blocking gates.

12.2 The five controls

CONTROL	HOW IT WORKS
Revision lock	A released drawing or BOM revision cannot be edited. A change creates a new revision through the approval path. A job records the revision it was built to.
Approval separation	Drafting, review, approval and release are different permissions. DSEW decides whether one person may hold more than one.
Quality gate	A failed or missing mandatory inspection or test blocks quality release. An NCR closes only after CAPA verification. Dispatch is blocked until release.
Immutable evidence	Measured values, photographs and signatures are stored as evidence that cannot be altered, with the user, time and device.
Closed means closed	When a job or order is closed, it is permanently read-only. Rework is a new linked record, so history is never rewritten.

12.3 Traceability

Each finished unit carries a QR identity. Scanning it opens the unit's record: the sales order, BOM revision, components with batch or heat numbers, route and operations, inspection and test results, and dispatch details. A forward trace answers "which units used this batch"; a backward trace answers "what went into this unit". An optional public verification page shows a controlled subset for the customer.

12.4 Stock integrity

Inventory is a ledger: every receipt, issue, transfer and adjustment is a posted entry. Corrections are made by reversal with a reason, never by silently changing a balance, so the quantity on hand can always be explained from its history.

SECTION 13

Artificial intelligence: options by hardware tier

DSEW asked that confidential client data must not leave its server. The AI design therefore runs on DSEW's premises, with capability that scales with the hardware DSEW chooses. DSEW can start with no extra hardware and add more after seeing real search and rules working on its own documents.

13.1 Tiers

TIER	WHAT IT IS	WHERE IT RUNS	WHAT IT DELIVERS
Tier 0	No model. Search and rules	The ERP VM; no extra hardware	Full-text search across documents and records; rule-based detection of missing documents and incomplete records; template and pattern extraction of drawing number, revision and PO number. Finds past quotations, POs, BOMs and jobs quickly and flags gaps
Tier 1	Small embedding model with a vector index	The ERP VM at 32 GB memory	Semantic search, for example "find the 100 kVA cast resin jobs", across the documents DSEW is permitted to see. Indexing runs in the background
Tier 2	Small language model on CPU	The ERP VM, background only	Overnight management summaries and batch tagging. Not offered for interactive drafting, where response time would be too slow
Tier 3	Local language model on a GPU machine	A separate small workstation with a 16 GB GPU on DSEW's LAN	Interactive drafting of quotations, work instructions, inspection plans and reports from approved templates, at usable speed. Verity calls it over the LAN, so data stays on the premises
Tier 4	External AI service	Off DSEW's server	Higher quality for non-confidential text only. Off by default; enabled only by DSEW's explicit decision; usage is charged by the provider

13.2 What is included and what is optional

- **Included with the Complete ERP (module 19):** Tier 0 and Tier 1, the integration of AI into the platform, and the documented use cases with their acceptance thresholds.
- **Optional, chosen by DSEW:** Tier 2 background summaries; Tier 3 GPU machine for interactive drafting; Tier 4 external service.
- **Not in the fee:** GPU hardware and any external service charges. Tier 3 hardware is priced after the use-case survey, so DSEW chooses it with real results in hand.
- The GPU machine is a **separate box**. Sharing a GPU with the Linux VM on the Windows host is not recommended, because it is fragile and complicates rollback.

13.3 Guardrails that apply at every tier

GUARDRAIL	SPECIFICATION
AI is never an approver	It does not approve quotations, release documents, alter records, authorise quality releases or act autonomously. No essential workflow depends on it
Inside the user's permissions	An AI feature acts as the calling user, with the same tenant, role and permissions, passing the same policy gate. It can never retrieve what the user could not open directly
Provenance	Results show the source records or documents they came from. Drafts are labelled and need a human to save and submit them
Acceptance thresholds	Each documented use case has defined inputs, outputs and an agreed threshold on DSEW's sample documents; it is accepted only if it meets the threshold
Data stays local	No document or record is sent off the server unless DSEW enables Tier 4
No training on DSEW data	Custom model training is excluded. DSEW's data is used only for retrieval at the moment of the request

13.4 Load control for AI

Indexing and any CPU-based model work run as background jobs with a concurrency limit and a daily window. They pause when interactive load rises, so AI can never slow the ERP for people working. Tier 3 inference runs on a different machine and adds no load to the ERP VM.

GO-LIVE GATE FOR AI

The selected AI tier must run within the latency agreed with DSEW, measured on the surveyed hardware, and each documented use case must meet its acceptance threshold on the agreed sample documents.

SECTION 14

Deployment topology and installation sequence

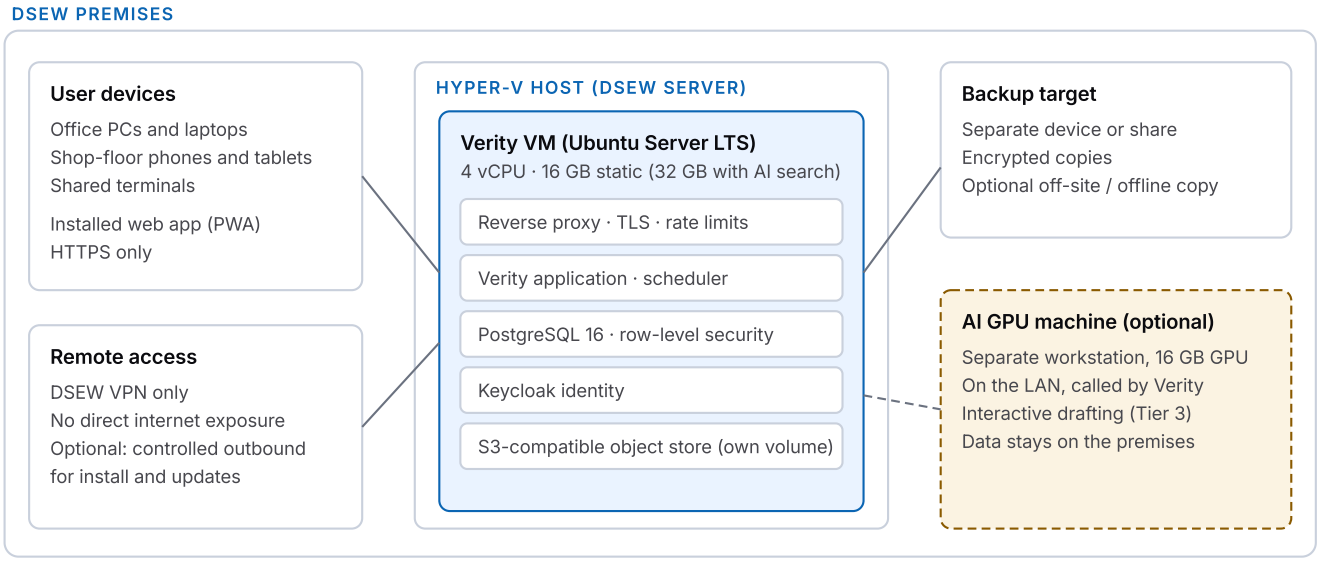


Figure 3. Deployment on DSEW's premises. The dashed AI machine is optional.

14.1 Installation sequence

#	STEP	OUTCOME
1	Pre-deployment compatibility check	Non-intrusive review of the host, virtualisation, network, certificate, backup target and policies. Pass or fail recorded per dependency; nothing on the host is changed
2	VM creation	Ubuntu Server LTS VM with the agreed static resources and volumes
3	Platform installation	Scripted install of the containers, database, identity provider and object store; secrets generated and sealed
4	HTTPS and DNS	Internal name and trusted certificate; PWA install verified from a real phone and a PC
5	Configuration	DSEW roles, numbering, workflows, custom fields, print templates and master-data import
6	Backup schedule	Jobs scheduled; first backup and restore rehearsal
7	Monitoring	Health view and alert thresholds set
8	Load and acceptance tests	Performance test against section 6.5 and UAT scripts, results recorded

14.2 Environments

A company-hosted test environment serves demonstrations and early hands-on use. The DSEW VM is installed at kickoff, used for UAT, and becomes production only after the gates in section 15 pass. A second production VM is not included.

SECTION 15

Acceptance tests and go-live gates

Every claim in this document is backed by a test that DSEW sees run and signs off. Production use begins only when all the gates pass, and the testing period starts on the date they pass and the handover is signed.

15.1 Go-live gates

#	GATE	EVIDENCE DSEW RECEIVES
1	Server survey and compatibility	Written pass/fail record per dependency
2	Installation and HTTPS on DSEW's host	Working access from office PCs and a real-phone upload; installed web app confirmed on supported devices
3	**Performance and resource acceptance**	Load test on DSEW's VM at 15 concurrent sessions against section 6.5; measured CPU, memory, disk and response times; resource footprint recorded
4	Combined backup and restore	Restore of database, files and identity configuration onto a clean environment; measured restore time
5	Outage behaviour	Identity and object-store outage tests: sessions continue, new sign-ins or uploads fail clearly, quality gates stay enforced
6	UAT scripts	Signed scenario results with date, tester and outcome
7	Data export reconciliation	Export reconciles with the system
8	Security checks	Tenant-isolation test, permission test, audit-trail immutability test, TLS configuration check
9	**AI latency and threshold** (if an AI tier is selected)	Selected tier runs within the agreed latency on the surveyed hardware; each documented use case meets its threshold on agreed samples

15.2 UAT scenarios

1. Enquiry to quotation to order to project.
2. BOM draft, review, approval and release; a released revision cannot be edited.
3. Material availability, purchase, receipt and issue to a job.
4. Manufacturing job with traveller; operations recorded; hold and resume.
5. Failed mandatory test blocks quality release and dispatch.
6. Dispatch released after approval; QR scan opens the unit record.
7. Data export, then backup and restore onto a clean environment.

15.3 Definition of done for every feature

Server-side authorisation, tenant-isolation test, audit entry, validation and error states, automated tests, documentation note, and demonstration in the review.

15.4 Defects and change control

SEVERITY	MEANING	EFFECT
Critical	Wrong data, a security issue, or work blocked with no workaround	Blocks go-live
High	Major function impaired	Blocks go-live
Medium	Function affected, workaround exists	Scheduled with DSEW
Low	Cosmetic or minor	Fixed in routine maintenance

DSEW has 5 business days after each demonstration or delivery to accept it or report material deviations. A shared issue list, visible from day one, shows severity, owner and status. Anything beyond the agreed scope follows a written change request with a revised cost or schedule, approved before work (documents 1 and 2).

SECTION 16

Requirements traceability matrix

Each of the 19 modules in DSEW's enquiry is mapped to the platform capability that carries it, the evidence that proves it, and the phase in which it is delivered under Option 3 (phases as in document 2).

#	ENQUIRY MODULE	PLATFORM CAPABILITIES USED	ACCEPTANCE EVIDENCE	PHASE
1	CRM, enquiry, quotation, sales order	Party, workflow, approval, numbering, revisions	Linked customer → enquiry → quotation → order; revision history; approval above threshold	1
2	Project management	Work, state, milestones, budgets	Project with milestones and budget versus actual; change recorded	1
3	Engineering / PLM	Entity revisions, release gate, ECR/ECO	Released revision locked; change via ECO; order records revision	1, 2
4	Document management	Numbering, controlled workflow, files, search	Numbered document through draft to issued; distribution history; search	1
5	Manufacturing execution	Work, state machine, operations, stock ledger	Job from BOM with material allocation and operation tracking	2
6	Travellers	Templates, checkpoints, QR	Printed traveller with QR; sign-offs recorded	3
7	Quality management	Inspection, evidence, gate, NCR/CAPA	Failed mandatory check blocks release; NCR closes after CAPA verification	3
8	Barcode, QR, traceability	Batch/heat/serial, genealogy, QR identity	Forward and backward trace for a unit and a batch	3
9	Procurement	Approval, supplier, PO, receipt	RFQ → comparison → PO → receipt; approval by value	2
10	Inventory	Stock ledger, reservation, count	Balances reconcile to ledger; reversal-only correction	2
11	Planning and MRP	BOM, demand, supply, rules	Shortage recommendations from a BOM-driven plan	2
12	Testing and equipment	Test templates, instruments, calibration	Test per product family with limits; report generated; calibration alerts	3
13	Finance and accounting	Ledger, invoices, payments, projections	Invoice with tax; payment reduces balance; receivables reconcile	4
14	HR and payroll	Party, timesheets, payroll rules	Sample month reproduces DSEW's own calculation	4

#	ENQUIRY MODULE	PLATFORM CAPABILITIES USED	ACCEPTANCE EVIDENCE	PHASE
15	Dispatch and logistics	Gate, packing, challan	Dispatch blocked on failed test; released after approval	3
16	Contracts and risk	Contract, obligations, alerts	Obligation and expiry alerts; LD exposure view	3
17	Dashboards and BI	Projections, permissions	Role dashboards reconcile to source records	4
18	Notifications and automation	Events, rules, scheduler	Trigger → escalation → notification demonstrated	4
19	AI use cases	Search, rules, embeddings, policy gate	Each use case meets its threshold on agreed samples	4

The MVP covers the nine modules in document 1, section 3, at the basic depth described there.

SECTION 17

Support, continuity and handover

17.1 Support during the testing period

Channels	team@plotarmour.in · +91 87440 69597
Hours	Business days, Monday to Saturday, 10:00 to 18:00 IST, remote
Acknowledgement	Within 1 business day
Defect plan	Fix or workaround plan for a reproducible defect within 3 business days
Included time	Up to 8 hours a month of testing assistance and maintenance
Escalation	Shubhankar Rakshit, CFO · team@plotarmour.in · +91 87440 69597
Covers	Defect warranty, testing and UAT assistance, routine maintenance (document 1, section 7)
Fee and duration	₹8,500 per month ex GST, billed monthly from go-live, for as many months as DSEW chooses

17.2 Continuity: if PlotArmour is unavailable

- **The system runs on DSEW's server** and keeps working day to day without PlotArmour.
- **Source code** of the DSEW-specific deliverables is in a repository DSEW controls from full payment.
- **Runbooks** let DSEW's IT team, or another developer, install, back up, restore, upgrade and roll back.
- **Open components** (PostgreSQL, Keycloak, Next.js and others) are standard technology that any competent engineer can maintain, and a licence inventory is provided.
- **Data portability:** a documented export is available at handover and on request during the testing period.
- Updates to Verity's own components come from PlotArmour; the terms are in document 3.

17.3 Handover package

- The working, tested application and the modules of the chosen option.
- Source code of the DSEW-specific deliverables.
- Architecture overview and database schema for the delivered scope.
- Deployment and configuration instructions, and the runbooks for backup, restore, upgrade and rollback.
- Administrator and user guides.
- Document templates and the workflow configuration register.
- API notes for what the scope uses.
- Load-test report, backup and restore report, UAT sign-off records and the security check results.
- Licence inventory of open components.

- Training for administrators and users.
- Credentials through a signed handover checklist, rotated at handover.

17.4 Training

Role-based sessions for administrators, engineering, stores, production, quality, finance and management, with a quick-reference guide for the shop floor. Training is delivered remotely unless on-site work is separately agreed.

SECTION 18

Risk register

#	RISK	IMPACT	MITIGATION	OWNER
1	The host cannot run the VM at the agreed resources	Schedule, cost	Pre-deployment compatibility check first; written alternative (second VM or small dedicated host) quoted before work proceeds	PlotArmour with DSEW IT
2	Server load higher than planned	Slow screens	Resource budget, background scheduling, load test at acceptance, health view; resources increased on the same architecture	PlotArmour
3	Other workloads on the host are affected	DSEW operations	Static VM allocation, separate disks, background work windows	DSEW IT, PlotArmour
4	Document volume larger than assumed	Storage	Separate volume sized from real samples; capacity report; alerts	DSEW, PlotArmour
5	Data loss or corruption	Business continuity	Nightly backups, encryption, restored and tested twice before go-live	PlotArmour, DSEW IT
6	Certificate or DNS not ready	Schedule	Raised at kickoff; internal CA option; tested early	DSEW IT
7	Delayed inputs (samples, decisions, formats)	Schedule	Named product owner; consolidated feedback; review windows; estimates revised for pending inputs	DSEW
8	Scope growth during delivery	Schedule, cost	Signed specification with module depth; written change requests	Both
9	AI results below expectation	Value	Tiered options, acceptance thresholds on DSEW samples, AI never essential to workflow, start with search and rules	PlotArmour
10	Confidential data exposure	Reputation	Local-only AI by default, isolation, permissions, audit, NDA	Both
11	User adoption on the shop floor	Value	Phone-first screens, role-based training, quick guide, early shop-floor trials	DSEW, PlotArmour
12	Dependence on the vendor after go-live	Continuity	Source handover, runbooks, open components, export (section 17.2)	PlotArmour

SECTION 19

Glossary

TERM	MEANING
PWA	Progressive Web App. A web application that can be installed on a device and opens like a native application, from one build
Tenant	The data-isolation boundary. DSEW's company data is one tenant
Row-level security	A PostgreSQL feature that makes the database itself filter every row by tenant, independent of application code
Command / Query	A request to change data, and a request to read it, handled by separate paths
Event	A recorded fact that something happened, which other modules can react to
Audit trail	The append-only record of every change
BOM	Bill of materials
ECR / ECO	Engineering change request / order
NCR / CAPA	Non-conformance report / corrective and preventive action
QAP / ITP	Quality assurance plan / inspection and test plan
MRP	Material requirements planning
RPO / RTO	Recovery point objective (data-loss window) / recovery time objective (time to restore)
vCPU	Virtual processor assigned to a virtual machine
Embedding / vector index	A numeric representation of text meaning and the index that finds similar items, used for semantic search
p95	The response time within which 95 % of requests complete
UAT	User acceptance testing
Hyper-V	Microsoft's virtualisation role in Windows Server, which hosts the Linux VM

DOCUMENT CONTROL

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